PHYS4055 Week 6 Tutorial Questions

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Question 1: Two atoms coupled to a common cavity mode

Two identical two-level atoms are resonantly coupled, with equal coupling strength g, to the same cavity mode. In the interaction picture, the Tavis–Cummings Hamiltonian is

$$\hat{H}_I = \hbar g \left(\hat{a}\hat{J}_+ + \hat{a}^\dagger \hat{J}_- \right),$$

where

$$\hat{J}_\pm = \hat{\sigma}_\pm^{(1)} + \hat{\sigma}_\pm^{(2)}.$$

The cavity initially contains no photons, atom 1 is excited and atom 2 is in its ground state:

$$|\Psi(0)\rangle = |e, g; 0\rangle.$$

Define the symmetric and antisymmetric single-excitation states

$$|S\rangle = \frac{|e,g\rangle + |g,e\rangle}{\sqrt{2}}, \qquad |A\rangle = \frac{|e,g\rangle - |g,e\rangle}{\sqrt{2}}.$$

1. Write the initial atomic state $|e, g\rangle$ as a superposition of $|S\rangle$ and $|A\rangle$.
2. Show that
$$\hat{J}_-|S\rangle = \sqrt{2}|g, g\rangle, \qquad \hat{J}_-|A\rangle = 0.$$
3. Explain why the symmetric state interacts with the cavity while the antisymmetric state does not.
4. The symmetric component evolves according to
$$|S; 0\rangle \longrightarrow \cos(\sqrt{2}gt)|S; 0\rangle - i \sin(\sqrt{2}gt)|g, g; 1\rangle,$$
while $|A; 0\rangle$ does not evolve. Write the complete atom–cavity state at time t.
5. Calculate the expected number of photons in the cavity,
$$\langle \hat{n}_{\text{cav}} \rangle = \langle \hat{a}^\dagger \hat{a} \rangle,$$
and the expected total number of excited atoms,
$$\langle \hat{N}_{\text{at}} \rangle = \left\langle \hat{\sigma}_+^{(1)} \hat{\sigma}_-^{(1)} + \hat{\sigma}_+^{(2)} \hat{\sigma}_-^{(2)} \right\rangle.$$
Verify that the total number of excitations is conserved.
6. What is the maximum probability of finding one photon in the cavity? Explain why the initial atomic excitation cannot be transferred completely to the cavity.
7. Calculate the probability that each atom is excited as a function of time. Show that, at an appropriate time, the excitation is transferred completely from atom 1 to atom 2.

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Question 2: Conditional filtering and state overlap

A source prepares one of the two non-orthogonal states

$$|\psi_\pm\rangle = \frac{2|0\rangle \pm |1\rangle}{\sqrt{5}}.$$

When a particular detector result occurs, the following conditional operation is applied:

$$\hat{K}_\mathrm{s} = \frac{1}{2}|0\rangle\langle 0| + |1\rangle\langle 1|.$$

The label "s" denotes the successful detector result.

1. Calculate the initial overlap $\langle\psi_+|\psi_-\rangle$. Can the two input states be distinguished perfectly?
2. Calculate the unnormalised output state $\hat{K}_\mathrm{s}|\psi_\pm\rangle$.
3. Find the probability of success for each input state.
4. Normalise the successful output states and calculate their overlap.
5. The operation has apparently transformed two non-orthogonal states into orthogonal states. Explain why this does not mean that a deterministic physical operation can always distinguish non-orthogonal states perfectly.
6. The other measurement result is represented by

   $$\hat{K}_\mathrm{f} = \frac{\sqrt{3}}{2}|0\rangle\langle 0|.$$

   Determine the conditional output when this result occurs and explain what information it provides about the input.